Block-Insure: Hybrid Evaluation of Authority-Aware Healthcare Insurance Adjudication

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***Abstract*—Healthcare claims combine confidential evidence, off-chain provider facts, probabilistic risk, human judgment, and funded settlement; no public dataset represents all five as ground truth. This paper presents Block-Insure, an Ethereum-compatible insurance protocol evaluated with 558,211 public-secondary Medicare claims plus eight controlled claim/registry attacks. Contracts enforce versioned policy economics, exact commit–reveal oracle consensus, deterministic auditor review, and pull-payment settlement, while AES-GCM evidence remains off-chain and Merkle roots bind external records. On 1,085 held-out providers, a calibrated Bernoulli model achieved ROC-AUC 0.8797 and PR-AUC 0.3576; logistic regression performed better at the selected operating point, limiting the claim to provider-risk prioritization. On 109,397 paired claims, combined rules and probabilistic routing detected all 54,705 injected attacks at 7.7881% untouched escalation. A 5,000-record runtime retained complete automatic and manual settlement workflows. Contract, reserve, evidence, gas, and load results expose both operational feasibility and bounded limitations. The contribution is a reproducible hybrid evaluation architecture, not proof of real-world claim-fraud detection.**

***Index Terms*—blockchain insurance, healthcare claims, hybrid dataset, oracle consensus, risk screening, confidential evidence**

I. Introduction

A healthcare claim is simultaneously an evidence, external-data, decision, and settlement problem. Blockchain can make state and money transitions auditable, but cannot independently prove that an off-chain medical event occurred—the reason authenticated and decentralized oracle mechanisms remain a separate trust layer [1], [2]. A model can prioritize attention, but cannot turn a weak provider label into confirmed claim fraud. Direct on-chain medical-document storage also creates confidentiality, cost, and permanence problems. Block-Insure therefore separates authority: contracts govern versions, roles, decisions, reservations, and balances; oracle workers report bound external observations; models advise routing; auditors resolve unsafe automation; and encrypted bytes remain off-chain.

Evaluation is equally fragmented. Prior healthcare-insurance systems have demonstrated blockchain adjudication and rule-based antifraud controls [3]–[6], while real Medicare studies show both the value and severe imbalance of provider-risk learning [7], [8]. Public claims provide realistic providers, beneficiaries, amounts, diagnoses, procedures, and time distributions, but not controlled truth for invoice reuse, altered amounts, registry mismatch, oracle conflict, stale versions, or proof corruption. Fully synthetic data provides those truths but weakens realism. This work combines both without conflating their labels.

The study asks whether (RQ1) real-derived aggregates rank weakly labeled providers under provider-held-out evaluation; (RQ2) explicit integrity rules and probabilistic screening complement each other on controlled attacks; (RQ3) a hybrid runtime preserves purchase-to-settlement functionality; and (RQ4) full-protocol evaluation reveals security, performance, evidence, and reserve boundaries hidden by model metrics alone.

The contributions are: (C1) a modular protocol for versioned policy economics, exact oracle consensus, confidential evidence, manual adjudication, appeals, and funded pull settlement; (C2) a provenance-preserving adapter for 558,211 public-secondary claims with distinct provider-risk, controlled-attack, and on-chain truth; (C3) provider-disjoint and paired-attack evaluations with calibration, comparators, ablations, and uncertainty; and (C4) contract, mutation, browser, Merkle/gas, evidence, reserve, and load experiments. Claims are bounded to the frozen local configuration; production decentralization, clinical truth, regulatory compliance, and universal fraud detection are outside scope.

II. Background and Related Work

Mackey *et al.* prototyped a blockchain antifraud framework for multi-stakeholder healthcare-claim adjudication, while identifying realistic empirical validation as future work [3]. Block-HI subsequently organized healthcare-insurance fraud into 12 scenarios and mapped them to blockchain controls, but did not learn from real claims or test an executable end-to-end settlement pipeline [4].

B-SAHIC implemented permissioned health-insurance processing with Hyperledger Fabric and measured transaction performance, but its predefined validation logic did not combine probabilistic risk, confidential evidence, and reserve-aware settlement [5]. ChainCare demonstrated an Ethereum/React insurance DApp and gas-oriented prototype evaluation; external-data/oracle validation remained outside its realized trust model [9]. Islam *et al.* combined Fabric, IPFS, anonymous credentials, and smart-contract fraud rules, while leaving data-driven detection as an extension [6].

For analytics, Kaur *et al.* combined explainable machine learning with blockchain-backed healthcare-fraud records, reporting strong ranking performance on large claims data [8]; Hancock *et al.* showed that imbalance handling and explainability materially affect Medicare provider-risk modelling [7]. Neither evaluates adversarial registry/oracle truth together with a complete claim lifecycle. FraudAuditor instead supports expert investigation of collusive health-insurance relationships, reinforcing the value of human review rather than forced autonomous classification [10].

Thus, existing work separately covers blockchain adjudication, fraud rules, provider-risk learning, and expert investigation. Block-Insure addresses the composition gap: it preserves distinct weak-label, controlled-attack, and protocol truth; binds oracle observations to versions; and evaluates model, evidence, contract, reserve, and browser behavior in one reproducible workflow.

III. System and Data Architecture

A. Authority and Lifecycle

The React client authenticates by wallet signature and performs evidence cryptography. Express orchestrates authenticated requests, projections, indexing, analytics, and notifications; MongoDB is not economic authority. Two workers independently reconstruct registry snapshots and submit version-bound observations, treating external data as an explicit oracle boundary [1], [2]. Seven Solidity modules divide protocol state: `InsuranceManager` controls claim transitions; `PolicyEconomics` snapshots rules and reserves coverage; `OracleCoordinator` enforces commit–reveal; `ClaimAdjudicator` owns assignment, votes, decisions, and the payout vault; `EvidenceRegistry` anchors identities and tree heads; benefits and deployment registries isolate auxiliary state.

A purchase snapshots terms and opens coverage. Submission validates ownership, incident, amount, documents, duplicate keys, and available coverage before reserving liability. An exact positive oracle quorum allocates payout; false, conflict, or timeout becomes eligible for review. Four auditors are assigned deterministically: three valid votes approve, two invalid votes reject, and expiry rejects. Insufficient treasury creates a visible `FUNDING_REQUIRED` liability; after funding, claimant withdrawal produces `SETTLED`. Appeals increment claim version so old oracle or decision records cannot finalize the new cycle.

B. Evidence, Registry, and Failure Containment

The browser encrypts with AES-GCM using claim/version/uploader/type as authenticated data [11], uploads ciphertext, and wraps the 256-bit key for proxy transformation; related PRE/blockchain work likewise separates protected health data from the ledger [12]. The backend need not learn plaintext or the key. A per-claim chain binds document order; an append-only RFC6962-style evidence tree supports inclusion and consistency proofs [13]; the chain anchors authenticated heads, not medical bytes, consistent with hybrid off-chain/on-chain storage principles [14]. Separately, oracle requests snapshot the healthcare-registry version/root, worker membership, quorum, and model identity. Missing model artifacts, disagreement, timeout, insufficient funds, unavailable storage, and stale projections fail without granting silent backend settlement authority.

TABLE I
Hybrid Partitions and Controlled Injection

Quantity	Train	Calibration	Test
Claims	343,537	105,277	109,397
Providers	3,420	905	1,085
Injected attacks	40,969	12,600	54,705
Injection rate	≈12%	≈12%	50%

C. Hybrid Dataset and Runtime

The implemented source is the compatible four-file Kaggle *Healthcare Provider Fraud Detection Analysis* Medicare benchmark [15]: provider labels, beneficiaries, inpatient claims, and outpatient claims joined only by native `Provider` and `BeneID`. It contains 558,211 claims (40,474 inpatient; 517,737 outpatient), 5,410 providers, and 138,556 beneficiaries. The public `PotentialFraud` field is a weak provider label, not claim truth; this interpretation follows provider-oriented Medicare fraud research rather than relabeling every claim [7].

For provider p and seed 20260828, the first 32 bits of $H(\texttt{split:}\|s\|p)$ assign 64%/16%/20% hash buckets to train/calibration/test. All claims of a provider remain together (Table I). Temporal features use only prior events; treatment amount ranges and 95th-percentile velocity thresholds are learned from training. Beneficiaries can cross partitions and are disclosed as a limitation.

Independent claim-ID hashes select eight attacks: used/cancelled invoice, inflated amount, date mismatch, blacklisted provider, near duplicate, provider compromise, and noisy record. Original and modified test rows remain paired by source claim. Generative augmentation has been studied for imbalanced healthcare-fraud learning [16]; our deterministic mutations instead test specified protocol failures and do not claim to reproduce a real fraud distribution. Thus *provider risk*, *controlled attack truth*, and *on-chain protocol truth* remain separate targets.

Offline studies use all applicable claims. Runtime uses an order-independent bottom-hash sample of 4,995 real-derived training records plus five stable normal/attack presets, producing 5,000 unique invoices and a depth-13 Merkle tree. The original claim amount is reimbursement plus deductible; demonstration value is $\max(amount/10{,}000, 0.0001)$, a local scale rather than currency conversion. A manifest records hashes, counts, seed, and version `blockinsure-hybrid-v1.2.0`.

IV. Methodology

The evaluation separates weak provider labels, injected claim-level irregularities, and executed protocol outcomes (Fig. 1); none substitutes for another.

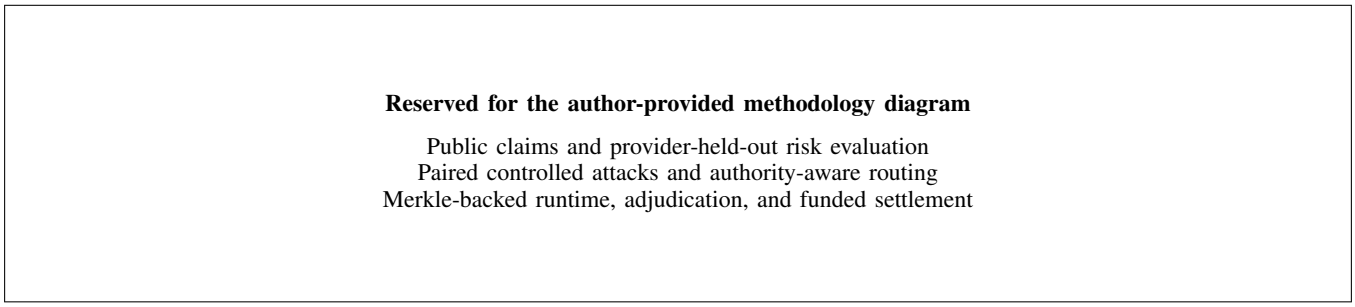


Fig. 1. Hybrid evaluation separates provider-level weak labels, paired controlled attacks, and a Merkle-backed runtime. Registry commitments establish inclusion, not clinical truth.

A. Provider-Risk Estimation

For provider p, nine aggregates describe claim/beneficiary volume, mean/maximum amount, inpatient/procedure/repeat ratios, mean chronic count, and activity-anomaly ratio. Training-only upper quartiles binarize these fields. Bernoulli Naive Bayes (BNB; Laplace $\alpha = 1$) estimates

$$\ell_y(\mathbf{x}) = \log \pi_y + \sum_j \left[x_j \log \theta_{jy} + (1 - x_j) \log(1 - \theta_{jy}) \right], \tag{1}$$

where π_y is class prevalence and θ_{jy} the smoothed feature probability. Softmax yields P_1; calibration providers fit a Platt sigmoid $\hat{p} = \sigma(a \operatorname{logit} P_1 + b)$ [17]. Calibration-only F_1 maximization selects $t^* = 0.35$ over $.01, \ldots, .99$; test labels never select thresholds. Comparators are majority negative, unweighted feature count, and unweighted/class-weighted logistic regression. Three leave-group-out ablations and 1,000 seeded provider bootstraps measure contribution and uncertainty [18].

B. Authority-Aware Claim Routing

The *separate* protocol model uses 18 claim/registry features, not provider-risk predictions. With missed-attack, false-escalation, and review costs $(C_m, C_f, C_r) = (10, 1, 0.25)$, cost-sensitive triage [19] uses

$$t_m = \max\left(\frac{C_r + C_f}{C_m + C_f}, t_{capacity} \right) = 0.217969. \tag{2}$$

The calibration-derived capacity threshold targets a nominal 20% model-review budget; mandatory integrity routing can exceed it. Hard comparison failure H and committed/fuzzy anomaly A determine

$$\mathcal{R}(\mathbf{x}, \hat{p}) = \begin{cases} \textsc{Reject-Oracle}, & H(\mathbf{x}) = 1, \\ \textsc{Manual-Review}, & A(\mathbf{x}) = 1 \ \vee \ \hat{p} \geq t_m, \\ \textsc{Auto-Verify}, & \text{otherwise.} \end{cases} \tag{3}$$

Probability can defer to review [20], not establish authoritative rejection. Both original and paired attack cohorts contain 109,397 rows; only the latter contains the 54,705 injected positives. Model-only, rule-only, and combined routing score the same paired inputs.

Algorithm 1: Version-bound adjudication.

1: Validate policy, claim, evidence, and available coverage.
2: Snapshot claim/registry versions, root, model, and workers.
3: Obtain version-bound commit–reveal observations.
4: Apply integrity checks and advisory routing above.
5: Matching quorum fixes the oracle verdict; otherwise enable review.
6: Review: four seats; approve with three valid votes; reject with two invalid votes or insufficient quorum at expiry.
7: Allocate liability if funded; otherwise mark funding required.
8: Settle on funded claimant withdrawal. Appeals restart verification under an incremented claim version.

C. Settlement, Reserve, and Evidence Experiments

For amount A, deductible rate/cap (d, D), and insurer share s (basis points), integer settlement is

$$\begin{aligned} q &= \min\big(A, \lfloor Ad/10^4 \rfloor, D\big), \\ L &= \lfloor (A - q)s/10^4 \rfloor, \qquad R = A - L, \end{aligned} \tag{4}$$

Here q is the deductible, L insurer liability, and R the patient share, subject to $reserved + settled + L \leq coverage$. Protected treasury is $U + B + \lfloor Er/10^4 \rfloor$ for unfunded liability U, buffer B, exposure E, and reserve ratio r.

Reserve experiments separate frequency and severity [21]: target $K \sim Poisson(Nf)$, $X_i = \exp(\ln m + \sigma Z_i)$, with N policies, frequency f, median m, spread σ, and standard-normal Z_i. The sampler uses a normal approximation when $Nf > 30$. Capped $A_i = \min(X_i, 1)$ produces contract liability L_i and annual loss $S = \sum_i L_i$. Across M periods, capital $C = Er$ gives

$$P_{short} = \frac{1}{M} \sum_j \mathbb{1}[S_j > C], \qquad ES = E[S - C \mid S > C]. \tag{5}$$

Evidence experiments measure AES-GCM, PRE capsule transformation, local storage, and Merkle proofs, plus eight tamper/availability/access failures. Contract and workflow measurements test executed behavior, not clinical truth.

D. Experimental Protocol and Reproducibility

The reported research snapshot is dated 28–29 August 2026, with contract hardening/Slither follow-up on 31 August; this manuscript's presentation revision is 21 October 2026. Figures

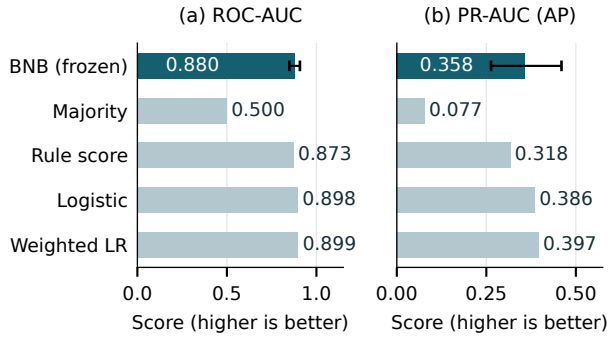


Fig. 2. Provider-held-out model comparison. Error bars are recorded 95% provider-bootstrap intervals for BNB only; none are available here for comparators. PR-AUC is average precision; positive weak-label prevalence is 7.74%.

replot recorded measurements, not newly executed experiments. Semantic identities bind datasets, models, code, compiler, and bytecode. Later application changes are not verified by these results.

Experiments used one local x64 Windows host, Node.js 24.16.0, Solidity 0.8.26, Cancun EVM, IR, one optimizer run, Hardhat 2.28.0, and chain 31337. Oracle quorum was exactly two, with 25-block commit/reveal windows. Manual review required four seats with 3-valid/2-invalid thresholds. Logistic comparators used 1,200 iterations, learning rate 0.08, and L2 penalty 0.002. Reserve experiments used 1,000 policies, 5,000 periods, seed 410034, 250 bootstrap resamples, and a 1% shortfall target; evidence used 30 repetitions.

Provider metrics use providers; attack metrics use paired claim rows. We report precision, recall, F_1, MCC, ROC-AUC, average-precision PR-AUC, Brier score, and ten-bin calibration error where available. PR-AUC is emphasized under imbalance [22]. Bootstrap intervals shown for BNB are not comparator significance tests. Local latency, throughput, gas, proof size, and bytecode remain environment-specific; load batches are not replicated capacity estimates. Slither 0.11.6 [23] used identical dependency/helper-filtered scope before and after hardening. Detector observations are not confirmed vulnerabilities.

V. Results

A. RQ1: Held-Out Provider Risk

Among 1,085 test providers, 84 (7.74%) carry the positive weak label. Frozen BNB produced TP=41, TN=914, FP=87, FN=43. ROC-AUC was 0.8797 (95% provider-bootstrap CI 0.8488–0.9078), PR-AUC 0.3576 (0.2631–0.4599), and F1 0.3868 (0.3011–0.4706). Majority accuracy was 92.26% despite zero recall: accuracy alone obscures screening failure.

Figure 2 and Table II show stronger logistic ranking and operating-point performance. BNB is a frozen, interpretable reference, not the best observed classifier. Removing volume/activity features caused the largest ranking loss (ΔROC=-0.1015, ΔPR=-0.1463). Removing coarse setting/clinical fields improved F1 by 0.0755, indicating encoding redundancy/noise, not clinical irrelevance.

TABLE II
Provider Operating Points (Weak Labels)

Model	Threshold	Precision	Recall	F1
Frozen BNB	.35	.3203	.4881	.3868
Majority	.50	0	0	0
Rule score	.55	.3047	.4643	.3679
Logistic	.28	.4000	.5476	.4623
Weighted logistic	.35	.3932	.5476	.4577

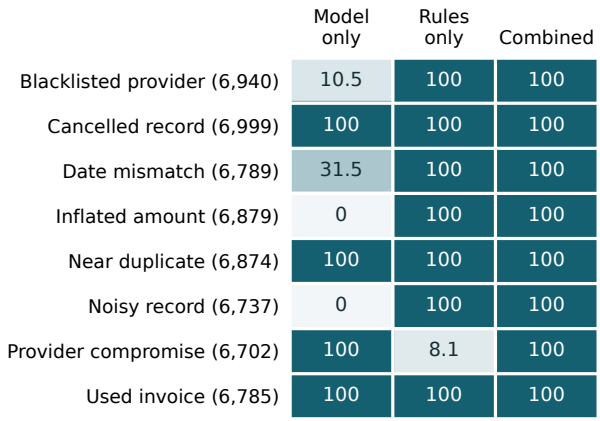


Fig. 3. Controlled-attack routing recall (%) across eight deterministic scenarios; parentheses give injected sample counts ($n = 54,705$ total). Positive routing means rejection of oracle verification or escalation, not a fraud conviction.

TABLE III
Routing Ablation: Original Controls and Paired Attacks

Strategy	Untouched routed (%)	Attack recall (%)	F1
Model only	.0037	55.25	.7117
Rules only	7.7845	88.74	.9034
Combined	7.7881	100.00	.9628

B. RQ2: Complementary Attack Routing

On the paired cohort, combined routing produced TP=54,705, TN=50,466, FP=4,226, FN=0. Figure 3 explains the complementarity: rules detect explicit integrity failures but only 8.09% of provider-compromise attacks; the model covers provider compromise but misses inflated and noisy records. Their union reaches every tested injection. This is bounded scenario coverage, not real-fraud accuracy.

Table III's untouched rate uses all 109,397 original controls, *not* the paired cohort's negative subset. Consequently, 7.7881% is not $FP/(FP + TN)$ from the paired confusion matrix. Combined routing adds less than 0.004 percentage points of original-cohort escalation above rules, while increasing injected-attack recall from 88.74% to 100%.

C. RQ3–RQ4: Executed Workflow and Operating Boundaries

Both recorded complete browser workflows reached `SETTLED`: automatic oracle processing and failed-oracle fallback resolved through three auditor approvals. Table IV separates successful execution from system limits. Root anchoring saved 99.62% gas against storing 500 record hashes, but this is a

TABLE IV
RECORDED LOCAL PROTOCOL MEASUREMENTS

Measurement	Result and boundary
Unit suites	Contract/backend/oracle 127/63/13; adapter 8 passed
Contract assurance	Statement/branch coverage 93.79/65.03%; sampled mutants killed 8/8
Browser workflows	Smoke/role 5/5; complete settlements 2/2 (about 84 s)
Registry/proofs	5,000 leaves, depth 13; 100/100 valid sampled proofs; mean verification .0246 ms
Registry storage gas	Root 43,851 vs. 500 hashes 11,402,190 gas
Evidence (1 MiB)	AES-GCM p95 encryption/decryption 3.432/3.341 ms
PRE key capsule	932 bytes; single-recipient p95 transformation/decryption 15.48/19.48 ms
Evidence adversarial cases	8/8 specified cases detected

TABLE V
LOCAL SUBMISSION BATCHES (NOT PUBLIC-CHAIN CAPACITY)

Concurrency	Successes/attempts	Successes/s	p95 (ms)
1	1/1	6.57	151.99
5	5/5	22.75	219.76
10	10/10	29.90	334.39
20	20/20	48.38	412.74
50	39/50	49.29	790.66

registry-storage comparison, not total insurance cost. Evidence timings measure local primitives, excluding browser and public storage/network latency.

At concurrency 50, throughput plateaued and 11/50 submissions failed (Table V); latency alone would conceal this limit. After the 31 August hardening, Slither observations fell from 108 to 95: high 6 to 4, medium 28 to 22, low 65 to 62, information 5 to 5, optimization 4 to 2. Manual triage did not identify an unresolved exploitable defect; retained observations and 65.03% branch coverage preclude an audit or security-proof claim. Hardened manager bytecode was 24,278 bytes, leaving 298 bytes below EIP-170 [24]. Public-network deployment/cost was not measured.

D. RQ4: Reserve Adequacy Under Stress

Figure 4 plots the recorded capital sweep. At 5% capital, shortfall probability was 0.22% at baseline (95% simulation-bootstrap CI 0.10–0.34%), 28.38% under heavy tails (27.34–29.43%), and 100% in the high-frequency and combined runs. The smallest *tested* ratios meeting the 1% target were respectively 5%, 8%, 10%, and 15%. They are grid recommendations under stated assumptions, not optimal or regulatory capital ratios. The result shows why one software reserve default is not robust across stress assumptions.

VI. DISCUSSION AND LIMITATIONS

A. Interpretation

RQ1 supports weak-label prioritization, not BNB superiority or autonomous fraud judgment: logistic comparators rank providers better, while BNB's operating-point recall and F1

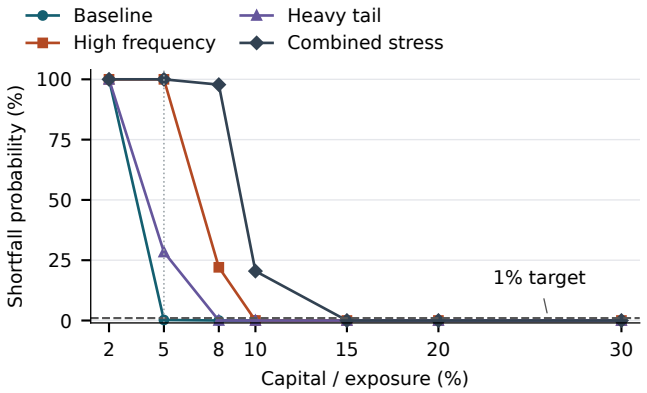


Fig. 4. Synthetic reserve stress: shortfall probability versus capital/exposure. Lines connect tested grid points, not a fitted model; dashed horizontal line marks the 1% target and dotted vertical line the configured 5% reserve. Each scenario uses 5,000 seeded periods.

remain moderate. RQ2 supports complementary routing on the specified injections, not unseen-fraud generalization. RQ3 is supported by automatic and auditor-routed settlements. RQ4 is supported by both positive and negative results: stress-reserve failures, local submission failures, bytecode pressure, incomplete branch coverage, and retained detector observations.

The contribution is the evaluated composition, not a new classifier or cryptographic primitive. Real-derived fields supply realistic distributions and weak provider risk; explicit mutations supply controlled attack truth; contracts establish versioned decisions and transfers. Probability can defer to review but cannot establish authoritative rejection. Exact oracle agreement permits automated processing; fallback requires the snapshotted auditor rules; claimant withdrawal requires a funded allocation. These mechanisms make authority inspectable without replacing institutional trust. Cross-paper accuracy rankings would be misleading across differing labels and splits; the within-study comparator and routing ablations provide the direct empirical comparisons.

B. Validity and Security Boundaries

Construct validity: `PotentialFraud` is inherited weak provider risk, not legal or claim-level fraud. The eight deterministic attacks exercise known fields and may be easier than adaptive behavior; 100% combined recall is therefore scenario coverage. Local settlement proves workflow execution, not clinical truth.

Internal validity: provider-disjoint splitting, training-only thresholds, calibration-only operating points, paired claims, and frozen seeds reduce leakage. However, beneficiaries cross partitions, Boolean features lose information, and comparators omit boosted, graph, and sequential models. Two oracle processes use separate accounts/snapshots but share one project operator; quorum collusion and admin-controlled registry publication remain threats.

External validity: the public-secondary Medicare-style source may not transfer across countries, insurers, coding

systems, or time. Only 5,000 records enter runtime; all 558,211 are used offline where applicable. Hardhat automining, local MongoDB/IPFS substitutes, one Windows host, and absent CPU/RAM capture prevent production throughput claims. Evidence timings exclude browser/network storage. Poisson-lognormal reserve scenarios omit dependence, inflation, expenses, premiums, reinsurance, and regulation.

Assurance/privacy: manifests bind source/output hashes, seeds, models, compiler, bytecode, and semantic results, but independent reproduction requires the same source files. Tests, 65.03% branch coverage, eight sampled mutants, and Slither triage are neither formal verification nor a professional audit; static analysis may still miss exploitable paths or composition defects. Formal model checking has been applied to blockchain claim-processing workflows and remains complementary future assurance [25]. Browser key storage is not production custody; revocation cannot retract recovered plaintext; access metadata and availability remain observable risks. No public-testnet deployment, independent hospital oracle, penetration/privacy assessment, HIPAA/GDPR certification, or statutory reserve claim is made.

VII. Conclusion

Block-Insure separates provider risk, controlled attack truth, and executed settlement. Frozen BNB achieved held-out ROC-AUC 0.8797 and PR-AUC 0.3576; combined routing covered all 54,705 injected attacks with 7.7881% original-cohort escalation. Two browser paths settled through version-bound oracle processing or auditor fallback. Reserve stress, local submission failures, incomplete coverage, and bytecode pressure bound these results. The contribution is an auditable evaluation architecture, not universal fraud detection or production insurance.

Future work prioritizes independent reproduction, beneficiary-disjoint evaluation, adaptive attacks, independently governed data sources, public-testnet measurements, stronger assurance, and qualified actuarial calibration.

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